

IT-Academy CRM — texnik topshiriqlar bajarilishi hisoboti

Uchta TZ (`prompt.md` , `prompt2.md` , `telegramBot.md`) bo'yicha holat va 2026-09-24 sessiyasida qo'shilgan o'zgarishlar.
Holat: `448916a` .

684 + 55 + 14

backend · frontend · E2E testlar (hammasi o'tadi)

24

commit shu sessiyada

11

migratsiya — hech biri DROP/DELETE emas

+14 935

qator kod, 137 fayl

1. Umumiy baho

Xulosa: uchala TZ ning asosiy talablari professional darajada bajarilgan. Har modul mavjud arxitektura ichida (servis qatlami, zod validatsiya, RBAC, audit, tranzaksiya), har bosqich to'liq test to'plami bilan yopilgan, hech bir mavjud test o'chirilmagan yoki bo'shashtirilmagan. Quyida **bajarilmagan yoki qisman** bandlar ochiq yozilgan — ular jami 15 ta, ko'pchiligi kichik hajmli.

Mezon	Holat	Dalil
Mavjud kodni buzmaslik	bajarildi	Har bosqichda 89–96 test fayli to'liq chopildi; API kontrakti o'zgarmadi (faqat maydon qo'shildi)
Migratsiya xavfsizligi	bajarildi	Har migratsiya oldidan <code>pg_dump</code> , <code>DROP/TRUNCATE/DELETE</code> 0 ta; vaqt mintaqasi xatosi (<code>now()</code>) topilib tuzatildi
Biznes mantiq bir joyda	bajarildi	Bot va kabinet bir xil servislarni chaqiradi; XP, qarz, davomat botda hisoblanmaydi
Xavfsizlik	bajarildi	Bog'lash kodi o'g'irlanishi, guruh chatidan bog'lanish, webhookning IP limiti — uchala nuqson sinovda topilib yopildi
Hujjatlashtirish	bajarildi	ACADEMY-2.0-AUDIT, TELEGRAM-BOT-AUDIT, telegram-architecture / security / deployment
To'liqlik	qisman	prompt.md: 3 band, prompt2.md: 2 band, telegramBot.md: 10 band — 7-bo'limga qarang

2. prompt.md — "Professional o'quv markaz tizimi" (75 band)

Bu TZ oldingi sessiyalarda bajarilgan; bu yerda holat koddan qayta tekshirildi (route, model, enum va sahifalar bo'yicha).

§	Talab	Holat	Koddagi asos / izoh
1–3	Arxitektura, Owner/Director dashboard, KPI kartalari		<code>dashboard.service</code> (bugun/oy bloklari), <code>ExecutivePage</code> , KPI havolalari

4–8	Davomat: bir klik, kalendar, statistika, reyting, o'qituvchi paneli	✓	attendance/stats , /ranking , /teacher-overview , kalendar endpointi; bot orqali ham
9	Avtomatik "kelmadi" bildirishnomasi + Telegram tayyorligi	✓	STUDENT_ABSENT_STREAK avtomatlashtirish, Telegram navbati
10–12	Gamifikatsiya: XP, daraja, qoidalar admin sozlaydi	✓	XpRule , Level — PUT /gamification/rules levels
13	Nishonlar; admin yangi nishon yaratishi	qisman	Nishonlar bor, tahrirlash va berish bor (PUT /badges/:id , POST /badges/award); yangi nishon yaratish endpointi yo'q
14	Leaderboard: davr, kurs, guruh filtrlari	✓	period , courseId , groupId
15–16	O'quvchi profili (tablar), streak	✓	studentProgress.service , Streak modeli, testlar
17–26	O'qituvchi moduli, 5 xil maosh, tasdiqlash va qulflash, tarix	✓	SalaryType (FIXED/PER_LESSON/PER_STUDENT/PERCENTAGE/MIXED), approve/unlock , davrlar
27–34	Moliya: tushum/xarajat, tranzaksiya, kassa/bank hisoblari, budjet	✓	TransactionType (INCOME/EXPENSE/TRANSFER/REFUND), AccountType (CASH...PAYME/UZUM), budjet vs fakt
35–38	Ota-ona, uy vazifasi, imtihon, progress	✓	Modellar va sahifalar; o'quvchi topshirishi shu sessiyada qo'shildi
39–42	O'qituvchi samaradorligi, retention, kurs/guruh analitikasi	✓	analytics , hisobotlar (retention , courses , groups)
43–44	Sotuv ↔ ta'lim bog'liqligi, global qidiruv	✓	Qidiruv: lead, o'quvchi, o'qituvchi, guruh, kurs, to'lov, tranzaksiya, sertifikat, ota-ona

45	Jadvallarda: qidiruv, filtr, saralash, sahifalash, ustun ko'rinishi , eksport	qisman	Hammasi bor, ustunlarni yashirish/ko'rsatish yo'q
46– 47	Eksport va hisobot markazi (13 tur)	✓	<code>/reports/:type/export</code> — 15 tur (sales...gamification), CSV/Excel
48– 49	Owner ruxsati, rolga qarab dashboard	✓	Dashboard bloklari ruxsatga qarab <code>null</code> keladi
50	Sozlamalar: markaz nomi, logo, telefon, manzil, ish vaqti, valyuta, o'quv yili...	✗	<code>Setting</code> modeli bor, modul sozlamalari o'z sahifalarida (ogohlantirish, gamifikatsiya, bildirishnoma). Umumiy "Markaz ma'lumotlari" sahifasi yo'q
51– 52	Audit ro'yxati, xavfsizlik (maosh faqat ruxsat bilan, audit o'chirilmaydi)	✓	XP/nishon auditi bor; <code>helmet</code> , <code>sameSite: strict</code> , RBAC
53– 57	Modellar, DECIMAL, unikal davomat, servis qatlami, VOID/REFUND	✓	<code>Decimal(14,2)</code> , <code>@@unique</code> , voiding, refund
58– 62	UI/UX, mobil, dark mode, holatlar, real ma'lumot	✓	PWA, mobil davomat, empty/loading/error, LocalStorage faqat UI
63– 64	Testlar, seed	✓	684 backend + 55 frontend + 14 E2E; <code>seed</code> va <code>perfSeed</code>
65	Unumdorlik (100k o'quvchi, 1M davomat)	✓	Indekslar, SQL agregatsiya; benchmark: 2 418 o'quvchi / 85k tranzaksiya / 150k bildirishnoma — hammasi <110 ms
66	Business overview: sana oralig'i Today / Week / Month / Last month / Year / Custom	qisman	Oy / chorak / yil + o'tgan davr bilan taqqoslash bor; "bugun", "hafta", "ixtiyoriy oralig" yo'q
67– 70	Foydalilik, ogohlantirishlar (8 tur), sotuv maqsadi, xodim samaradorligi	✓	<code>AlertType</code> 17 tur (LOW_GROUP_CAPACITY, SALES_TARGET_ACHIEVED ham), <code>targets</code>

71–75	Kelajakka tayyor arxitektura, bosqichlar, sifat tekshiruvi	✓	Telegram, onlayn to'lov, kabinet, AI — hammasi keyin qo'shildi va ishladi
-------	--	---	---

3. prompt2.md — "Academy CRM 2.0" (36 talab)

§	Talab	Holat	Izoh
1	Audit	✓	docs/ACADEMY-2.0-AUDIT.md , A-O formatida
2–3	Student lifecycle, at-risk (health_score, 4 daraja)	✓	6 signal, fon vazifasi, sabablar
4–5	O'quvchi va ota-ona kabineti	✓	Bu sessiyada: darslar, dastur, yutuqlar, sertifikatlar, vazifa topshirish API
6	Telegram bot (student/parent/teacher/manager/owner), sozlamalar, navbat/retry	✓	To'liq bot (4-bo'lim), turxkanal sozlamalari, navbat 5 urinish
7	Onlayn to'lov abstraksiyasi, webhook, imzo, idempotentlik	✓	Bu sessiyada: PaymentIntent , provayder interfeysi, sandbox, HMAC; Click/Payme kaliti kelganda 1 fayl
8	Xonalar, jadval, to'qnashuv	✓	Room (sig'im, jihozlar), 4 shartli to'qnashuv
9	Kurrikulum	✓	Modul → mavzu, progress
10	Imtihon dvigateli: savollar bazasi, qiyinlik, tasodifiy savollar , vaqt, urinishlar, qayta topshirish, avto/qo'lda baholash	qisman	Bank, qiyinlik, mavzu, vaqt va urinishlar (bu sessiyada), avtomatik/qo'lda baholash bor. Tasodifiy tanlash bor (imtihonga birlashtirishda: soni + mavzu + qiyinlik, examAttempt.service.ts). Yo'q: har o'quvchiga alohida variant, blueprint, savol tartibini aralashtirish
11	Sertifikat + PDF + QR tekshiruv	✓	Bu sessiyada chop etish / PDF sahifasi
12–16	Lead skoring, avtomatik taqsimot, voronka, referral, chegirma	✓	DiscountType 7 tur (FAMILY... FIRST_PAYMENT), stacking, chegara
17–19	HR, o'qituvchi analitikasi, fikr/NPS	✓	Maxfiy ma'lumot ruxsati, anonim fikr, salbiy fikr xabari
20	Inventar (ko'chirish bilan)	✓	Bu sessiyada filiallararo atomik ko'chirish qo'shildi (avval bir tomonlama edi)
21	Multi-branch	✓	branchAccess , o'qishdagi izolyatsiya

22–24	AI yordamchi, owner dashboard, avtomatlashtirish	✓	14 xavfsiz tool; 7 trigger
25	Global qidiruv (invoice, sertifikat, telefon, ID)	✓	Sertifikat, telefon, kod prefiksi bor; "invoice" — to'lov kvitansiyasi sifatida
26	Bildirishnoma markazi: filtrlar, turlar INFO/SUCCESS/WARNING/CRITICAL	qisman	Filtrlar va muhimlik (LOW/NORMAL/HIGH) bu sessiyada qo'shildi. TZ dagi 4 nomli tur ogohlantirishlarda bor (severity), bildirishnomada 3 darajali priority — ma'no jihatidan bajarilgan, nomlari farq qiladi
27	Mobile-first / PWA	✓	
28	Xavfsizlik (16 band, jumladan backup verification)	✓	Bu sessiyada <code>verify-backup.sh</code> — nusxa vaqtinchalik bazaga tiklanadi
29–30	Audit 2.0 (before/after), unumdorlik	✓	Benchmark qayta chopildi
31–33	Testlar, baza, UX	✓	
34–36	Bosqichlar, mavjudni buzmaslik, natija formati	✓	

4. telegramBot.md — "Telegram Bot Platform" (70 bo'lim, 15 bosqich)

Arxitektura qarori (siz tasdiqlagan): bot **CRM backend ichida modul** (Variant A). TZ §1 dagi "alohida servis → REST API" diagrammasidan farq qiladi, lekin sababi aniq: markazda hisobi yo'q ota-onalar bor — alohida servis ular nomidan API ga murojaat qilish uchun keng vakolatli tokenga muhtoj bo'lardi (TZ §31 ga zid). Biznes mantiq takrorlanmaydi (§58, §60).

§	Talab	Holat	Izoh
2–3	Audit (17 bo'limli format)	✓	<code>docs/TELEGRAM-BOT-AUDIT.md</code> ; TZ dagi 6 jadvaldan 4 tasi dublikat bo'lardi — yaratilmadi
5	Ro'l tizimi — CRM ruxsatlaridan	✓	Menyu ruxsatga qarab quriladi; xodim uchun haqiqiy <code>AuthUser</code>
6	Bog'lash: qisqa muddatli, bir martalik, replay, brute-force	✓	15 daqiqa, <code>codeUsedAt</code> , 5 xato → 15 daqiqa blok. Jiddiy nuqson topilib yopildi (kod boshqa chatga o'tkazilardi)
7	Yangi jadvallar	✓	<code>TelegramSession</code> , <code>TelegramEvent</code> , <code>TelegramBroadcast</code> ; qolganlari mavjud jadvallar
8–10	Student menyu, profil, davomat kalendari	✓	10 tugma; kalendar oylik, oldingi oyga o'tish

11	Vazifalar: ro'yxat, tafsilot, topshirish (matn/rasm/hujjat)	✓	Matn, rasm, PDF. DOCX ataylab qabul qilinmaydi (hujjatlar bilan bir xil qoida). CRM'da ham yo'q edi — API yaratildi
12	Imtihonlar; Telegramdan imtihon boshlash	qisman	Natijalar bor; onlayn imtihonni botdan boshlash yo'q (savollar bilan ishlash kabinetda)
13–14	XP, to'lovlar (👉 To'lash tugmasi)	✓	To'lov tugmasi provayder holatini ko'rsatadi, bot to'lovni tasdiqlamaydi
15	Ota-ona: bir nechta farzand, tanlash, izolyatsiya	✓	Tanlov sessiyada, doiradan tashqari id rad etiladi
16–18	O'qituvchi: guruhlar, bugungi darslar, tezkor davomat; XP/streak CRM'da	✓	Hamma "keldi" bilan boshlanadi, 2–3 bosish
19	Vazifa berish (title, description, deadline, attachment)	qisman	Sarlavha, muddat, tavsif bor; botdan fayl biriktirish yo'q
20–21	Sotuv: leadlar, hot, follow-up, status; amallar: Call, Contact, Follow-up, Trial	qisman	Ro'yxatlar, kartochka, status (LOST sabab bilan), follow-up bajarish bor. Botdan qo'ng'iroq yozish va follow-up yaratish yo'q ; WON ataylab CRM'da
22	Owner menyu (13 band: tushum... KPI, marketing, hisobotlar, AI)	qisman	Ko'rsatkichlar, qarzdorlar, at-risk, ogohlantirishlar, AI bor. Teacher KPI, Marketing, Hisobotlar bo'limlari yo'q
23	Kunlik avtomatik hisobot	✓	Mavjud <code>DAILY_DIGEST</code> Telegramga ham boradi (format TZ namunasidan farq qiladi)
24	Bildirishnoma turlari (student/parent/teacher/sales/owner)	✓	19 tur; bu sessiyada +5 (vazifa, imtihon, daraja, sertifikat)
25	Sozlamalar (yoqish/o'chirish) — bot ichida	qisman	Turxkanal sozlamalari CRM va kabinetda bor; botda ⚙ Sozlamalar bo'limi yo'q
26	Navbat, retry, xato jurnali	✓	Mavjud <code>NotificationDelivery</code>
27–29	Buyruqlar, inline tugmalar, sahifalash	✓	17 buyruq, menyu <code>setMyCommands</code> bilan sinxron
30	Qidiruv (bot ichida)	✗	Botda qidiruv yo'q (CRM da bor)
31–33	Xavfsizlik, callback tekshiruvi, audit	✓	<code>docs/telegram-security.md</code> — har band testga bog'langan
34	Broadcast: auditoriya, preview, navbat, statistika; rasm/hujjat/tugma	qisman	6 auditoriya, preview, navbat, statistika bor; faqat matn
35–40	Filial doirasi, to'lov arxitekturası, sertifikat, referral, at-risk, AI	✓	
41, 44	Alohida <code>apps/telegram-bot</code> loyiha va konteyner	boshqacha	Variant A — modul, alohida konteyner kerak emas (yuqoridagi izoh)

42– 43, 45– 46	ENV, webhook, logging, xato ishlovi	✓	Polling (sinov) + webhook (production) skriptlari; token logga tushmaydi
47– 51	UX, bo'sh holatlar, API dizayni, javob standarti, validatsiya	✓	
52	Testlar: unit, integratsion, xavfsizlik, E2E , yuklama	qisman	~110 integratsion/xavfsizlik testi, yuklama skripti bor; Playwright E2E Telegram uchun qo'llanmaydi (webhook orqali integratsion test shu vazifani bajaradi)
53– 55	Unumdorlik, rate limit, monitoring	✓	300–700 update/s; GET /api/telegram/health
62	7 ta alohida hujjat (architecture, api, auth, permissions, notifications, deployment, testing)	qisman	4 hujjat: audit, architecture, security, deployment — api/auth/permissions/notifications/testing mazmuni shular ichida, alohida fayl emas
56, 63– 70	15 bosqich, yakuniy natijalar, ish tartibi	✓	Barcha 15 bosqich bajarildi va kommit qilindi

5. Xavfsizlik: sinovda topilgan va yopilgan nuqsonlar

Nuqson	Oqibati	Yechim
Bog'lash kodi muddatsiz, <code>verifiedAt</code> tekshirilmasdi	Kodni ko'rgan har kim bildirishnomalarni o'ziga burardi	Muddat, bir martalik, blok, audit (<code>238f3c0</code>)
<code>/uzish</code> matni bosiladigan havola edi	Bir tap bilan tasdiqsiz uzilardi (haqiqiy sinovda ro'y berdi)	Ikki qadamli tasdiq (<code>11b5fa4</code>)
Bot guruh chatida ishlardi	Guruhdan <code>/start <kod></code> — qarz va davomat butun guruhga	Faqat shaxsiy chat (<code>c953c10</code>)
Webhook umumiy IP limitiga tushardi	Telegram bitta IP — 30 tugmadan keyin bot to'xtardi (yuklama testi topdi)	Alohida <code>webhookLimiter</code> (<code>c953c10</code>)
Migratsiyada <code>now()</code> — Asia/Tashkent bazasi	Kodlar 5 soat kelajakka qo'yilardi	Aniq o'tmish sanasi; qoida hujjatga (<code>238f3c0</code>)
Ombor ko'chirish bir tomonlama	Tovar hisobdan yo'qolishi mumkin edi	Atomik tranzaksiya (<code>e1ef6de</code>)
Yangi ruxsatlar productionga tushmasdi	Yangi modullar 403 berardi	<code>db:sync-permissions</code> deploy'da (<code>bfe6126</code>)

6. Bu sessiyada qo'shilgan o'zgarishlar

6.1. CRM (PHASE 14, 8 qism)

Kommit	Nima	Baza / API	Test
b02e868	Onlayn to'lov arxitekturasini: provayder interfeysi, HMAC imzo, idempotent webhook, sandbox provayder, kvitansiya mavjud <code>paymentService</code> orqali	<code>PaymentIntent ; /payments/online/* , /payments/webhook/:provider</code>	+7
c2d1933	Kabinet: kelgusi darslar, kurs dasturi, yutuqlar, sertifikatlar, o'qituvchi	<code>/portal/lessons , /curriculum , /certificates</code>	+2
d17e5e6	Sertifikatni chop etish / PDF (A4, QR), chop etishda CRM menyusi yashirinadi	<code>GET /certificates/:id (egalik bilan)</code>	+2
132ead9	Imtihon vaqti (1–600 daqiqa) va urinishlar (0–20), <code>EXPIRED</code> holati, urinishni boshlash	<code>Exam.durationMinutes/maxAttempts ; /exams/:id/attempts/:studentId/start</code>	+3
e1ef6de	Filiallararo atomik ko'chirish; qabul filialida mahsulot o'zi ochiladi; oddiy harakatda TRANSFER rad etiladi	<code>POST /products/transfers</code>	+5
096ce03	Bildirishnoma muhimligi (LOW/NORMAL/HIGH) va shaxsiy sozlamalar (tur x ilova/Telegram); SYSTEM o'chirilmaydi	<code>Notification.priority , NotificationSetting ; /notifications/settings</code>	+5
4a2df24	<code>verify-backup.sh</code> — nusxa vaqtinchalik bazaga tiklanadi va tekshiriladi; benchmark qayta chopildi	—	skript sinovi
0dbf73d	Bot buyruqlari: <code>/qarz</code> , <code>/darslar</code> , <code>/davomat</code> , <code>/holat</code> , <code>/uzish</code> ; menyu sinxron	—	+6

6.2. Telegram bot (15 bosqich)

Kommit	Bosqich	Nima	Test
79acde9	0	Audit — 17 bo'lim; jiddiy nuqson topildi	—
f92d233	1	Poydevor: router, sessiya, inline tugmalar, chat chegarasi, hodisa jurnali, xato ishlovi	+17

f308e51 , 35880cb	1	Polling (sinov), telegram:check , telegram:webhook	+4
238f3c0	2	Bog'lash xavfsizligi	+11
5c5dc1a	3–4	O'quvchi/ota-ona: 8 bo'lim, vazifa topshirish (matn/rasm/PDF), farzand tanlash; kabinet API 7 endpoint	+16
77aeebe	5	O'qituvchi: guruhlar, bugungi darslar, tezkor davomat, vazifa berish	+9
0784fcd	6–7	Sotuv (leadlar, hot, status, follow-up) va rahbar (ko'rsatkichlar, qarzdorlar, at-risk, ogohlantirishlar)	+6
a52594d	8	5 yangi bildirishnoma turi, bir tranzaksiyada, bitta servis	+6
f69225a	9	Broadcast: 6 auditoriya, preview, navbat, statistika; ruxsat broadcast.send	+5
3b8ece7	10–12	To'lov tugmasi, do'st taklifi, AI yordamchi	+5
c953c10	13–15	Xavfsizlik auditi, yuklama testi, monitoring, production hujjati	+3

6.3. Yangi jadvallar va ustunlar

- **Jadvallar:** payment_intents , notification_settings , telegram_sessions , telegram_events , telegram_broadcasts
- **Ustunlar:** exams.durationMinutes/maxAttempts , notifications.priority , telegram_links.codeExpiresAt/codeUsedAt/telegramUserId/lastSeenAt , homework_submissions.answerText , notification_deliveries.broadcastId
- **Enum'lar:** AttemptStatus.EXPIRED , NotificationPriority , TelegramEventStatus , BroadcastAudience , NotificationType +5
- **Ruxsat:** broadcast.send (90-ruxsat, deploy'da sinxronlanadi)

6.4. Yangi skriptlar va hujjatlar

- scripts/verify-backup.sh , backend/scripts/telegramCheck.ts , telegramWebhook.ts , telegramLoad.ts
- docs/TELEGRAM-BOT-AUDIT.md , telegram-architecture.md , telegram-security.md , telegram-deployment.md ; ACADEMY-2.0-AUDIT.md va DEPLOYMENT.md yangilandi

7. Qilinmagan yoki qisman — bir joyda

#	Band	Nima yetishmaydi	Hajm	Muhimlik
1	prompt.md §50	Umumiy "Markaz ma'lumotlari" sozlamalar sahifasi (nom, logo, telefon, manzil, ish vaqti, valyuta, o'quv yili)	o'rta	o'rta
2	prompt.md §13	Admin yangi nishon yaratishi (hozir faqat tahrirlash/berish)	kichik	past
3	prompt.md §45	Jadvallarda ustun ko'rinishini boshqarish	kichik	past

4	prompt.md §66	Business overview: "bugun", "hafta", "o'tgan oy", "ixtiyoriy oraliq"	kichik	past
5	prompt2.md §10	Har o'quvchiga alohida tasodifiy variant (tanlashning o'zi bor)	o'rta	o'rta
6	prompt2.md §26	Bildirishnoma turi nomlari (INFO/SUCCESS/WARNING/CRITICAL) — ma'no bajarilgan, nom farq qiladi	—	hujjatlashtirilgan
7	telegramBot.md §12	Onlayn imtihonni botdan boshlash	katta	o'rta
8	telegramBot.md §19	O'qituvchi botdan vazifaga fayl biriktirishi	kichik	past
9	telegramBot.md §20	Botdan qo'ng'iroq yozish va follow-up yaratish	o'rta	o'rta
10	telegramBot.md §22	Owner botida Teacher KPI, Marketing, Hisobotlar	o'rta	past
11	telegramBot.md §25	Botda  Sozlamalar (bildirishnoma yoqish/o'chirish) — CRM va kabinetda bor	kichik	past
12	telegramBot.md §30	Bot ichida qidiruv	o'rta	past
13	telegramBot.md §34	Broadcast rasm/hujjat/tugma bilan	o'rta	o'rta
14	telegramBot.md §62	api/auth/permissions/notifications/testing — alohida hujjat fayllari	kichik	past
15	telegramBot.md §36, prompt2 §7	Click/Payme haqiqiy provayderi — merchant kalitlari sizdan	kichik (1 fayl)	sizga bog'liq

Bulardan hech biri mavjud funksiyaga xalaqit bermaydi. 1, 5, 7, 9, 13 — foydali kengaytmalar; qolganlari qulaylik.

8. Production uchun sizdan kerak

- `.env.production` : `TELEGRAM_BOT_TOKEN` , `TELEGRAM_BOT_USERNAME` , `TELEGRAM_WEBHOOK_SECRET` (kompyuterdagi bilan boshqa), `TELEGRAM_POLLING=false` .
- Deploy → `npm run telegram:webhook -- https://<domen>` → `npm run telegram:check` .
- Kompyuterda polling ishlab tursa, u production webhook'ini o'chiradi — serverga chiqargach kompyuterda `false` qiling.
- Click/Payme merchant kalitlari — `backend/src/services/payments/` ga bitta provayder fayli.
- Bot tokeni chatda bir marta ochilib qolgan edi — `/revoke` qilinganiga ishonch hosil qiling.

Hisobot 2026-09-24 da tayyorlandi. Testlar: `npm test --workspace backend` (684), `npm test --workspace frontend` (55), `npm run test:e2e` (14). Yuklama: `npm run telegram:load --workspace backend` .